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Graphene synthesis project update for previous week

Objective 1: Production of 300 g of biochar

MB3051
Objective: Test longer milling (4x45 sec)
Experiment details
18.3 g biochar (MB3040, KFT 4.3%) milled (Blendtec, 4x45 sec, active cooling between milling intervals) with 27.5 g KOH, unloaded in glove box -> 41.5 g brown powder, no change in appearance during bottling
Rotating oven, 800 °C, 1 h, 3 °C/min, 40.8 g powder used, 7.2 g (90 ml) output
Result: Typical yield, mostly species 1
Conclusion: No apparent benefit of longer milling
Recommended action: Use standard 90 sec milling time for further experiments

MRa394
Objective: 1:1 test with air-dried biochar
Experiment details
22.2 g biochar (MB3045, KFT 5.1%, air-dried) milled (Blendtec, 90 sec) with 22.2 g KOH, unloaded in glove box -> 42.8 g brown powder, no change in appearance during bottling
Rotating oven, 800 °C, 1 h, 3 °C/min, 42.8 g powder used, 7.2 g (40 ml) output
Result: Low yield (34%), mostly species 1, small pores, compact product
Conclusion: Yield for 1:1 experiments inconsistent, effect of air-drying unclear
Recommended action: Focus on 1:1.5 experiments

MB3054
Objective: 1:1 test with 2% wet biochar
Experiment details
19.6 g biochar (MB3034, KFT 2.0%) milled (Blendtec, 90 sec) with 19.6 g KOH, unloaded in glove box -> 36.1 g brown powder, no change in appearance during bottling
Rotating oven, 800 °C, 1 h, 3 °C/min, 32.0 g powder used, 7.5 g (70 ml) output
Result: Normal yield (47%), species 1 + atypical structures, material unusually voluminous (9 ml/g)
Conclusion: Conditions not suitable for production,
Recommended action: Focus on 1:1.5 experiments

MRa398A+B

Objective: 1:1.5 test with increasing batch sizes

Experiment details

57.2 g biochar (MB3016/3018/3020 combined, KFT 4.6%) milled (Blendtec, 2x45 sec) with 85.8 g KOH, unloaded in glove box -> 140.2 g dark brown powder, color change from light brown to dark brown during milling (consequence of excessive heating?), no further change in appearance during bottling

A: Rotating oven, 800 °C, 1 h, 3 °C/min, 64.1 g powder used, 12.6 g (110 ml) output

B: Rotating oven, 800 °C, 1 h, 3 °C/min, 76.0 g powder used, 14.2 g (90 ml) output

Result: Normal yield (49%, 47%), mostly species 1, lower amount of atypical structures or species 2 compared to earlier experiments without color change

Conclusion: Quality acceptable, batch size increase for milling leading to color change (due to excessive heating?; beneficial?), reaction batch size unproblematic and further scale-up possible

Recommended action: Continue scale increase, investigate cause and effect of color change

MB3055

Objective: 1:1.5 test with increasing batch sizes

Experiment details

87.2 g biochar (various combined lab batches, KFT 4.7%) milled (Blendtec, 2x45 sec) with 130.8 g KOH, unloaded in glove box -> 214.2 g dark brown powder, color change from light brown to dark brown during milling (consequence of excessive heating?), no further change in appearance during bottling

Rotating oven, 800 °C, 1 h, 3 °C/min, 101.6 g powder used, 19.4 g (105 ml) output

Result: Normal yield (48%), mostly species 1, lower amount of atypical structures or species 2 compared to earlier experiments without color change

Conclusion: Quality acceptable, batch size increase for milling leading to color change (due to excessive heating?; beneficial?), reaction batch size unproblematic and further scale-up possible

Recommended action: Continue scale increase, investigate cause and effect of color change

Overall status

- 17 rotating oven batches completed, total 100 g (mostly pure) and 42 g (less pure, mixtures) product
- max batch size not reached
- Performance of 1:1 experiments on large scale unsatisfactory
- product quality consistency moderate, blender performance and biochar lot variability suspected cause for quality and consistency deficits
- product density variable (8-15 ml/g), down to 5 ml/g for 1:1 ratio
- current frequency is four batches (approx. 80 g product) per week, output can likely be increased to >20 g per batch
- step 1 production on lab scale stopped, first pilot plant batch is finished, second batch in drying
- So far, products only characterized by SEM, MRa389A and C sent for conductivity measurement

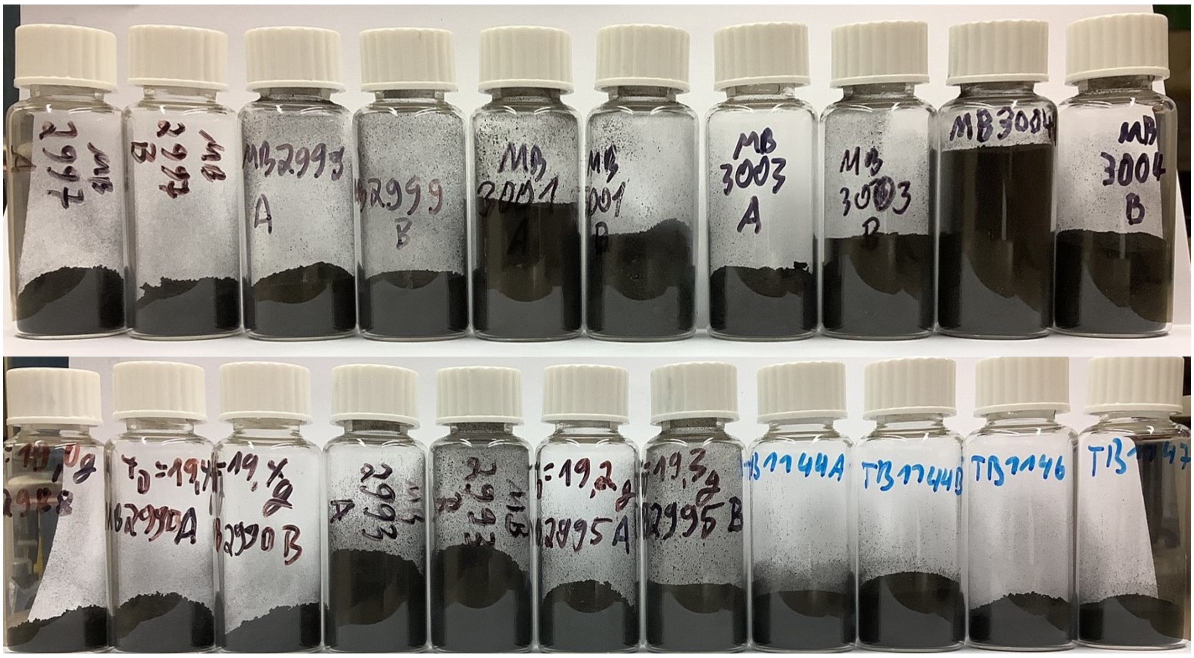
Objective 2: Testing/Optimization on small scale

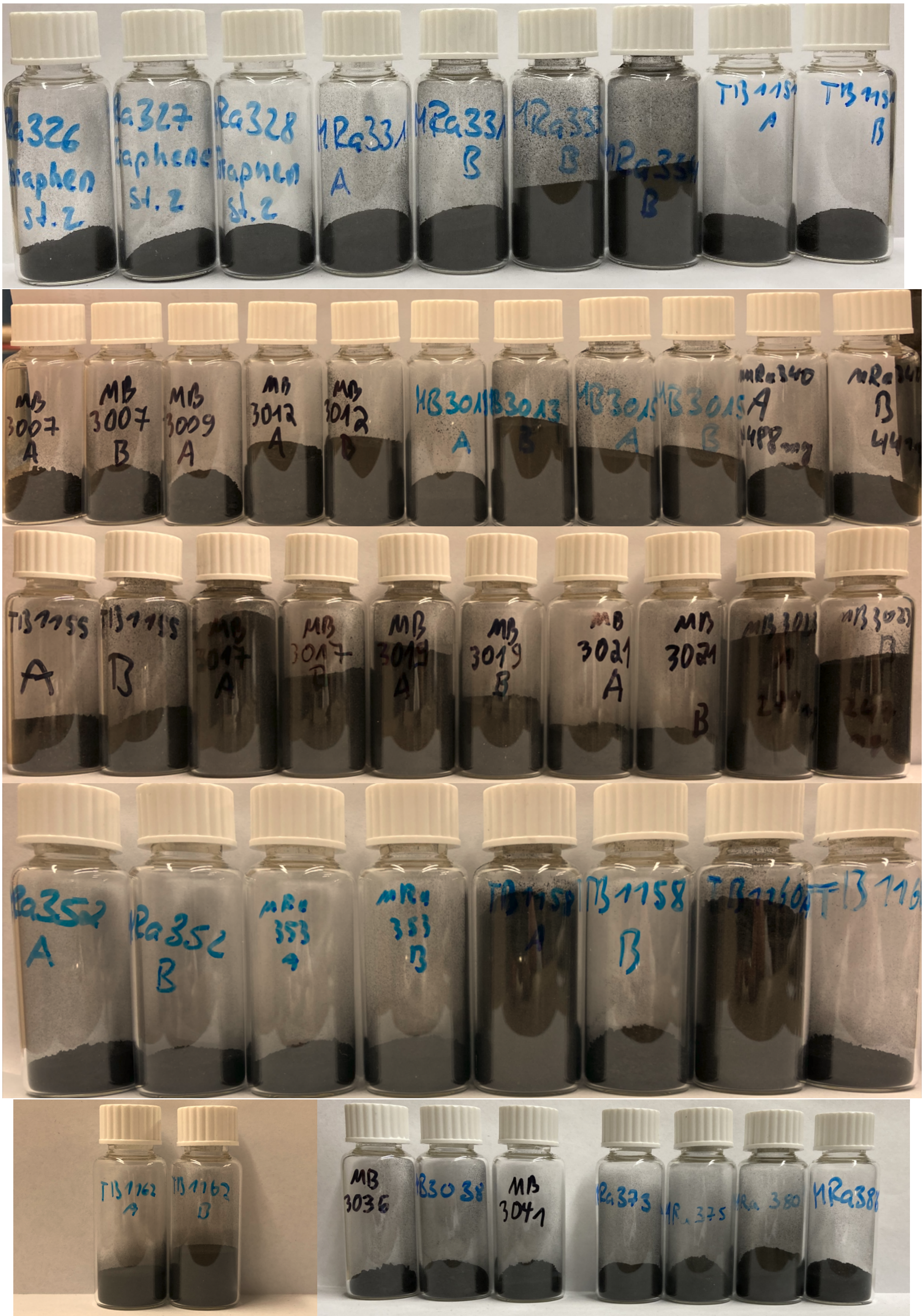
MRa395
Objective: 1:1 test with 2% wet biochar as comparison to MB3054
Experiment details
19.6 g biochar (MB3034, KFT 2.0%) milled (Blendtec, 90 sec) with 19.6 g KOH, unloaded in glove box -> 36.1 g brown powder, no change in appearance during bottling
800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A), 601 mg (3.0 ml) output
Result: Normal yield (46%), higher degree of species 1 (compared to MB3054), product inhomogeneous, uneven pore size distribution, compact
Conclusion: Conditions not suitable for production, poor performance due to poor milling or low moisture content of biochar
Recommended action: Focus on 1:1.5 experiments

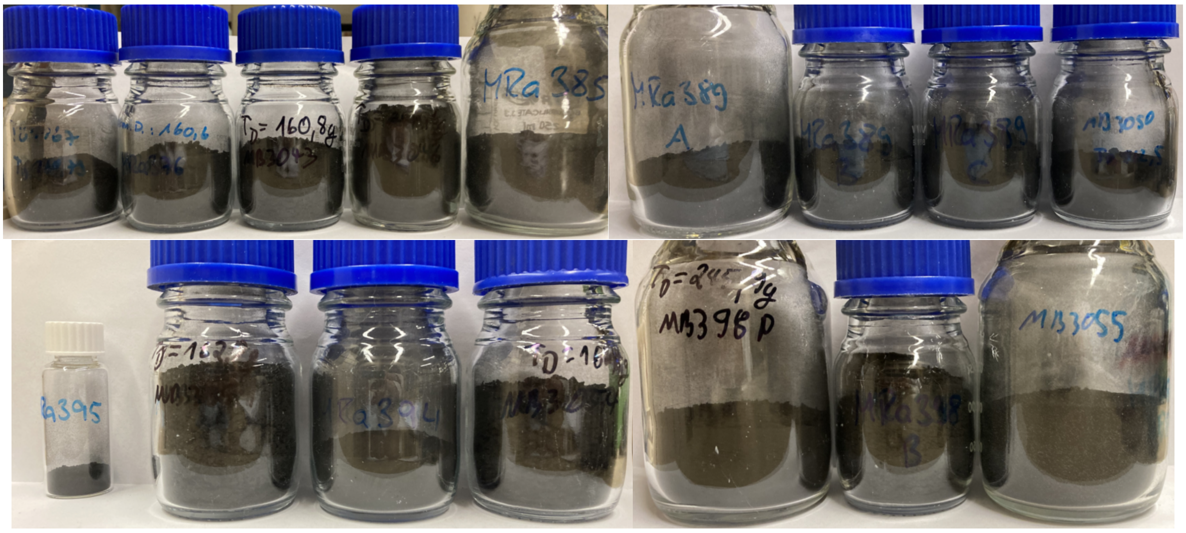
No new experiments on small scale were done this week

Overall status
- Latest changes (increased water content, 1:1 ratio biochar/KOH) not suitable or beneficial

Product pictures




MRa331 A
MRa331 B
MB 3007 A
MB 3007 B
MB 3009 A
MB 3012 A
TB1155 A
TB1155 B
MB 3019 B
MB 3021 A
MB 3021 B
MRa 353 B
TB1158 B
TB1162 A
TB1162 B
MB 3036
MB 3041


T_D = 160,8g
MRa 385
MRa 389
A
MB398 P
MB3055

Analytical data

- SEM (and partially Raman) results available on sharepoint for
    - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323, TB1141, MRa326, MRa327, MRa328, MB2990A, MB2990B, MRa329, MRa331A, MRa331B, MRa333A, MRa333B, MRa334A, MRa334B, MRa2993A, MRa2993B, MB2995A, MB2995B, MB2997A, MB2997B, TB1144A, MB1144B, MB2999A, MB2999B, TB1146, TB1147, MB3001A, MB3001B, MB3003A, MB3003B, MB3004A, MB3004B, TB1151A, TB1151B, MB3007A, MB3007B, MB3009, MB3013A, MB3013B, MB3015A, MB3015B, MRa340A, MRa340B, MB3017A, MB3017B, MB3019A, MB3019B, MB3021A, MB3021B, MB3023A, MB3023B, MRa352A, MRa352B, MRa353A, MRa353B, TB1158A, TB1158B, TB1142, TB1160A, TB1160B, TB1162A, TB1162B, MB3026, MB3027, MB3028, MB3030, TB1165, TB1166, MB3036, MB3038, MRa373, TB1167, MRa375, MB3041, MRa376, MB3043, MRa380, MRa385, MRa386, MB3046, MRa389A, MRa389B, MRa389C, MB3050, MB3051, MB3054, MRa394, MRa395, MRa398A, MRa398B, MB3055 (step 2 products)
    - 900561-500MG, 2 lots (Sigma-Aldrich reference)
    - 924458-500MG (Sigma-Aldrich reference “Single-layer graphene sheets for battery, Bio-sourced”)
    - Graphene oxide

- TEM results available for
    - MB2955A2 (step 2 product)
    - MB2976A (step 2 product)
    - TB1137-1 (step 2 product)

- BET results

Sample	Multipoint BET Area [m^2/g]	Langmuir Surface Area [m^2/g]	Species
MB2976A	1.88 x 10^3	2.55 x 10^3	mostly 1
MRa329	1.76 x 10^3	2.39 x 10^3	2
MRa333A	2.09 x 10^3	5.74 x 10^3	2
MRa334A	1.86 x 10^3	2.50 x 10^3	mostly 2
MB3001A	1.613 x 10^3	7.763 x 10^3	mostly 2
MB3004A	1.491 x 10^3	2.032 x 10^3	mostly 2
MB3001A/3004A	1.471 x 10^3	1.983 x 10^3	mostly 2
924458-500MG	1.112 x 10^3	1.421 x 10^3	n.a.
900561-500MG	1.942 x 10^1	2.442 x 10^1	n.a.
TB1142	1.257 x 10^3	1.684 x 10^3	1
TB1160B	1.240 x 10^3	1.677 x 10^3	1
TB1165	1.520 x 10^3	2.020 x 10^3	1

Experiment overview – Step 1

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output	
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.4 bar	742 g	Water	2.6 g	
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.5 bar	647 g	Water	1.2 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	9.5 bar	500 g	Water	1.3 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	10.6 bar	589 g	Water	0.8 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	->	10.5 bar	500 g	Water	0.9 g	Combined and homogenized as lot no MRa231
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	11.0 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	->	10.5 bar	500 g	Water	0.8 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	10.9 bar	569 g	Water	1.6 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	->	13.2 bar	577 g	Water	1.4 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.6 bar	555 g	Water	1.5 g	
Comment	very low amount of foreign material observed and removed during filtration															Combined and homogenzied as lot no MB2946X
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	->	13.5 bar	578 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	->	12.7 bar	611 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	->	9.7 bar	593 g	Water	1.8 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.8 bar	6 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	->	12.6 bar	475 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output	Drying	KFT
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar	->	10.2 bar	538 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h		->	11.1 bar	534 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar	->	11.2 bar	575 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar	->	12.8 bar	488 g	Water	1.7 g		
Comment	some foreign material observed and removed during filtration																
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar	->	10.4 bar	558 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	12.5 bar	548 g	Water	1.9 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar	->	8.5 bar	589 g	Water	1.5 g		
Comment																	
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar	->	8.7 bar	562 g	Water	1.8 g		
Comment																	
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h				3957 g	Water	14.9 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar	->	10.5 bar	300 g	Water	0.6 g		
Comment	Reaction performed with stirring																
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar	->	16.1 bar	563 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	8.8 bar	310 g + 603 g	Water	1.3 g		
Comment	Hemp boiled with 1% H_2SO_4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid																
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h				4112 g	Water	15.4 g		
Comment																	
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M	Sulfuric acid	185 °C	24 h				4283 g	Water	16.5 g	40 °C	4.9%
Comment																100 °C	2.3%
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4384 g	Water	15.8 g	40 °C	3.7%
Comment																	
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4258 g	Water	14.9 g	40 °C	5.3%
Comment																	
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4232 g	Water	16.6 g	40 °C	5.5%
Comment																100 °C	1.7%

Experiment	Reactor	Raw material			Acid			T	t	pressure	Wash		Output	Drying	KFT
MB2985	AV5	79.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	200 °C	24 h		4241 g	Water	15.4 g	40 °C	3.7%
Comment														100 °C	1.9%
MB2986	AV5	79.9 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4128 g	Water	16.0 g	60 °C	2.7%
Comment														100 °C	1.7%
MB2989	AV5	80.2 g	Canadian Rockies Hemp	999.3 g	1.00%	0.10 M	Sulfuric acid	180 °C	23 h		4314 g	Water	18.2 g	40 °C	5.9%
Comment															
MB2991	AV5	81.3 g	Canadian Rockies Hemp	999.3 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4223 g	Water	19.1 g	40 °C	5.7%
Comment															
MB2992	AV5	81.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	160 °C	23.5 h		4262 g	Water	19.5 g	40 °C	6.2%
Comment															
MB2994	AV5	80.8 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4159 g	Water	20.2 g	40 °C	5.2%
Comment															
MB2996	AV5	82.7 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4210 g	Water	20.0 g	40 °C	4.6%
Comment															
MB2998	AV5	84.0 g	Canadian Rockies Hemp	999.6 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h		4263 g	Water	19.8 g	40 °C	5.0%
Comment	Incuded nickel and steel material samples for corrosion/cleaning tests														
MB3000	AV5	82.5 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4431 g	Water	19.8 g	40 °C	5.0%
Comment															
MB3002	AV5	80.0 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	190 °C	25 h		4245 g	Water	18.5 g	40 °C	2.0%
Comment															
MB3005	AV5	81.4 g	Canadian Rockies Hemp	999.4 g	1.00%	0.10 M	Sulfuric acid	200 °C	24 h		4280 g	Water	19.0 g	40 °C	2.0%
Comment															
MB3006	AV5	80.8 g	Canadian Rockies Hemp	1000.1 g	1.00%	0.10 M	Sulfuric acid	190 °C	23 h		4182 g	Water	17.4 g	40 °C	2.1%
Comment															
MB3008	AV5	82.2 g	Canadian Rockies Hemp	1000.2 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4238 g	Water	18.0 g	40 °C	2.6%
Comment															
MB3010	AV5	81.3 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4235 g	Water	18.9 g	40 °C	2.0%
Comment															
MB3011	AV5	82.4 g	Canadian Rockies Hemp	999.4 g	1.00%	0.10 M	Sulfuric acid	160 °C	24 h		4244 g	Water	23.5 g	40 °C	2.2%
Comment															
MB3014	AV5	82.8 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	160 °C	23 h		1460 g	Water	23.6 g	40 °C	1.3%
Comment															
MRa341	AV5	82.3 g	Canadian Rockies Hemp	1000.4 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		1232 g	Water	19.2 g	40 °C	2.1%
Comment															

Experiment	Reactor	Raw material		Acid				T	t	pressure	Wash		Output	Drying	KFT
MB3016	AV5	81.7 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	185 °C	21 h		4232 g	Water	19.7 g	40 °C	2.1%
Comment															
MB3018	AV5	81.2 g	Canadian Rockies Hemp	1000.1 g	1.00%	0.10 M	Sulfuric acid	195 °C	22 h		4266 g	Water	19.0 g	40 °C	2.1%
Comment															
MB3020	AV5	81.1 g	Canadian Rockies Hemp	1000.2 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4278 g	Water	18.9 g	40 °C	1.7%
Comment															
MB3022	AV5	81.8 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4262 g	Water	19.0 g	40 °C	2.9%
Comment															
MB3024	AV5	80.7 g	Canadian Rockies Hemp	999.6 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4256 g	Water	18.4 g	40 °C	2.0%
Comment															
MRa349	AV5	80.8 g	Canadian Rockies Hemp	999.4 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4254 g	Water	19.1 g	40 °C	2.2%
Comment															
MRa354	AV5	80.2 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4414 g	Water	18.4 g	40 °C	1.7%
Comment															
MRa355	AV5	80.2 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4262 g	Water	18.2 g	40 °C	1.8%
Comment															
MRa358	AV5	80.8 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		3317 g	Water	18.8 g	40 °C	2.1%
Comment															
MB3025	AV5	81.4 g	Canadian Rockies Hemp	1000.5 g	1.00%	0.10 M	Sulfuric acid	165 °C	21.5 h		4229 g	Water	18.4 g	40 °C	1.1%
Comment															
MB3029	AV5	81.6 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4178 g	Water	18.8 g	40 °C	1.7%
Comment															
MB3031	AV5	82.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	190 °C	25 h		4178 g	Water	19.8 g	40 °C	1.5%
Comment															
MB3032	AV5	81.9 g	Canadian Rockies Hemp	1000.5 g	1.00%	0.10 M	Sulfuric acid	180 °C	23.5 h		4341 g	Water	19.0 g	rt	4.8%
Comment															
MRa367	AV5	81.9 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24.5 h		4256 g	Water	19.8 g	rt	4.7%
Comment															
MRa368	AV5	81.4 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	185 °C	27.5 h		4290 g	Water	20.3 g	rt	5.5%
Comment															
MB3034	AV5	84.7 g	Canadian Rockies Hemp	1000.2 g	1.00%	0.10 M	Sulfuric acid	180 °C	24.5 h		4254 g	Water	20.0 g	40 °C	2.0%
Comment															
MB3037	AV5	80.7 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	185 °C	21.5 h		4325 g	Water	18.6 g	40 °C	1.7%
Comment															

Experiment	Reactor	Raw material			Acid			T	t	pressure	Wash		Output	Drying	KFT
MB3039	AV5	81.2 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h		4253 g	Water	19.1 g	40 °C	1.8%
Comment															
MB3040	AV5	80.0 g	Canadian Rockies Hemp	1002.2 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4301 g	Water	18.8 g	40 °C	
Comment														rt	4.3%
MRa3042	AV5	80.0 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	185 °C	26 h		4026 g	Water	19.3 g	40 °C	
Comment														rt	4.3%
MB3044	AV5	82.4 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h		4316 g	Water	20.2 g	40 °C	
Comment														rt	4.5%
MB3045	AV5	82.8 g	Canadian Rockies Hemp	1000.1 g	1.00%	0.10 M	Sulfuric acid	160 °C	23 h		3964 g	Water	22.8 g	rt	5.1%
Comment															
MB3047	AV5	81.1 g	Canadian Rockies Hemp	1000.1 g	1.00%	0.10 M	Sulfuric acid	160 °C	24 h		3836 g	Water	22.7 g	rt	5.3%
Comment															
MB3048	AV5	80.2 g	Canadian Rockies Hemp	1000.2 g	1.00%	0.10 M	Sulfuric acid	160 °C	23 h		4275 g	Water	23.9 g	rt	3.6%
Comment															
MB3049	AV5	80.7 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	155 °C	25 h		4248 g	Water	23.3 g	rt	3.6%
Comment															

Comments

- Red text for T and t shows experiments where the temperature tracks have not yet been evaluated (or are not available)

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
TB1131	A	1.2 g	MB2933	1.2 g	KOH	(90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										(+ water)		(atm. Pressure)		
TB1132	A	1.3 g	MB2935	1.3 g	KOH	(90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder										(+ water)		(atm. Pressure)		
TB1133	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1134	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2946	A	1.0 g	MB2946X	1.0 g	KOH	(90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2947A	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded				
Comment		ground biochar: brown powder; see temperature trend														
MB2947D	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g	10% HCl	100 °C	N2 stream	38 mg
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-1	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2952C2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2955A	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	7 mg
Comment		Quartz tube shattered during experiment										(+ water)		(atm. Pressure)		
MB2955A2	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1136	A	0.26 g	MB2946X	0.39 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	14 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962A	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962B	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963A	A	0.22 g	MB2960	0.22 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963B	A	0.29 g	MB2960	0.29 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-1	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-2	A	0.28 g	MB2946X	1.13 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	23 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-3	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2965A	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B2	B	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2966A	B	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2966B	A	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967A	B	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967B	A	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970A	B	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970B	A	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971A	B	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971B	A	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972A	B	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972B	A	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2975A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2975B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2976A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2976B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2979A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2979B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2980A	B	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980B	A	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980C	B	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980D	A	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0165A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0165B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.15 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.11 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
KJo-0167A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0167B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g	10% HCl	100 °C	N2 stream	0.06 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.02 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0169	B	0.80 g	MB2978	1.20 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0170	B	1.07 g	MB2978	1.61 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g	10% HCl	100 °C	N2 stream	0.62 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0171A	B	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0171B	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172A	B	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.24 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172B	A	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.17 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0175	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										(+ water)		(atm. Pressure)		
KJo-0176	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0177	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Volume	Species	Appearance
TB1138	A	1.54 g	various	2.31 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	59 g 10% HCl	100 °C N2 stream	0.59 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	38%			
MRa320	B	1.43 g	various	2.14 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl	100 °C N2 stream	0.55 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	38%			
MB2984	B	1.40 g	various	2.10 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	53 g 10% HCl	100 °C N2 stream	0.48 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	34%			
TB1139	A	1.39 g	KJo-0173	2.08 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl	100 °C N2 stream	0.54 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	39%			
TB1140A	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	49 g 10% HCl	100 °C N2 stream	0.46 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	36%			
TB1140B	B	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	50 g 10% HCl	100 °C N2 stream	0.48 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	38%			
MRa323	A	1.06 g	KJo-0173	1.60 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.44 g		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	41%			
TB1141	A	1.08 g	various	1.62 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	30 g 10% HCl	100 °C N2 stream	0.40 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	37%			
MRa326	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.51 g		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	39%			
MRa327	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%			
MB2988	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.42 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	40%			
MRa328	B	1.20 g	KJo-0173	1.80 g KOH (90%)	mill (3 min)		N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.53 g		1	black/grey brittle
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%			
MRa329	A	1.04 g	KJo-0173 (extra dried)	1.56 g KOH (90%)	mill (1 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.43 g		2	shiny grey voluminous
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	42%			
MB2990A	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.50 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	39%			
MB2990B	B	1.04 g	MB2982	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.42 g		1	black/grey brittle
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	40%			

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa331A	B	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.64 g		Mostly 1	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	50%			voluminous
MRa331B	A	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g		Mostly 1	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%			voluminous
MRa333A	A	1.28 g	MB2985 (extra dried)	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.21 g		2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	17%			voluminous
MRa333B	B	1.04 g	MB2985 (extra dried)	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.26 g		2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	25%			voluminous
MRa334A	A	1.04 g	MB2985 (extra dried)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.48 g		Mostly 2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	46%			voluminous
MRa334B	B	0.88 g	MB2985 (extra dried)	1.06 g KOH (90%) 0.26 g NaOH (98%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	33 g 10% HCl	100 °C N2 stream	0.39 g		Mostly 2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	45%			voluminous
MB2993A	A	1.28 g	MB2982	1.15 g KOH (90%) 0.77 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.22 g		Mostly 2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	18%			voluminous
MB2993B	B	1.04 g	MB2982	0.94 g KOH (90%) 0.62 g NaOH (98%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.19 g		Mostly 2	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	18%			voluminous
MB2995A	A	1.28 g	MB2983 (extra dried)	1.92 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.54 g	3.9 ml	Mostly 1	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	42%	7.3 ml/g		voluminous
MB2995B	B	1.04 g	MB2983 (extra dried)	1.56 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g	5.6 ml	1/2 mix	shiny grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%	12.3 ml/g		voluminous
TB1144A	A	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	38 g 10% HCl	100 °C N2 stream	0.61 g	5.0 ml	1 + fibres	black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	59%	8.2 ml/g		somewhat shiny
TB1144B	B	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (15 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	54 g 10% HCl	100 °C N2 stream	0.36 g	6.0 ml	2 (?)	black/grey shiny
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	28%	16.9 ml/g		
MB2997A	A	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.55 g	3.6 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	6.6 ml/g		somewhat shiny
MB2997B	B	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)	mill (5 min)	no	N2	3 °C/min	800 °C	1 h	40 g 10% HCl	100 °C N2 stream	0.45 g	2.9 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	6.4 ml/g		somewhat shiny
MB2999A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.68 g	4.6 ml	Mostly	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	53%	6.8 ml/g	1/2 mix	voluminous

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB2999B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (25 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.40 g	4.6 ml	1 + fibres	shiny black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	38%	11.5 ml/g		voluminous
TB1146	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)		N2	3 °C/min	800 °C	1 h	51 g 10% HCl	100 °C N2 stream	0.52 g	4.0 ml	1	black/grey
		ground biochar/KOH mix (brown powder, MB2990) dried at 100 °C, then compacted to two pellets of equal size									(+ water)	(atm. Pressure)	41%	7.6 ml/g		somewhat shiny
TB1147	B	1.08 g	MB2983 (extra dried)	1.62 g KOH (90%)	mill (4 min)		N2	3 °C/min	800 °C	1 h	37 g 10% HCl	100 °C N2 stream	0.36 g	5.0 ml	Mostly 1	black/grey
		ground biochar/KOH mix (brown powder, MB2995) dried at 100 °C, then compacted to two pellets of equal size									(+ water)	(atm. Pressure)	33%	14.1 ml/g		somewhat shiny
MB3001A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.49 g	10.6 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	38%	21.5 ml/g		very voluminous
MB3001B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.48 g	8.2 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	46%	17.2 ml/g		very voluminous
MB3003A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.57 g	5.7 ml	Mostly 1	black/grey
		freshly ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	45%	9.9 ml/g		somewhat shiny
MB3003B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.44 g	9.0 ml	Mostly	shiny black/grey
		freshly ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	42%	20.7 ml/g	1/2 mix	voluminous
MB3004A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.59 g	17.4 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	46%	29.4 ml/g		very voluminous
MB3004B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)		N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.41 g	8.4 ml	Mostly	shiny black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	39%	20.7 ml/g	1/2 mix	voluminous
TB1151A	A	1.40 g	KJo-0174	2.10 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	51 g 10% HCl	100 °C N2 stream	0.53 g	2.5 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	38%	4.7 ml/g		brittle
TB1151B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (15 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.45 g	2.5 ml	1	black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	42%	5.6 ml/g		brittle
MB3007A	A	1.20 g	KJo-0174	1.80 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.50 g	3.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	42%	6.0 ml/g		barely shiny
MB3007B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (15 Hz, 30 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.43 g	2.9 ml	1	black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	42%	6.7 ml/g		barely shiny
MB3009	A	1.28 g	MB2992	1.92 g KOH (90%)	mill (1 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.41 g	3.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	32%	7.4 ml/g		brittle
MB3012A	A	1.28 g	MB2983 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.64 g	10.2 ml	1/2 mix	shiny black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	50%	16.0 ml/g		voluminous

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Experiment	Oven	Qty	Lot	Base						grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash		Drying		Output	Vol/Dens	Species	Appearance
MB3012B	B	1.04 g	MB2983 (2% water)	1.56 g	KOH	(90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.44 g	10.8 ml	Mostly	shiny black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		42%	24.6 ml/g	1/2 mix	voluminous
MB3013A	A	1.28 g	MB2983 (2% water)	1.92 g	KOH	(90%)				ball mill (35 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.58 g	4.3 ml	Mostly	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		45%	7.5 ml/g	1/2 mix	
MB3013B	B	1.04 g	MB2983 (2% water)	1.56 g	KOH	(90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.40 g	12.8 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		38%	32.2 ml/g		very voluminous
MB3015A	A	1.20 g	MB2986 (2% water)	1.80 g	KOH	(90%)				ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	45 g	10% HCl	100 °C	N2 stream	0.39 g	8.0 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		33%	20.5 ml/g		voluminous
MB3015B	B	1.00 g	MB2986 (2% water)	1.50 g	KOH	(90%)				ball mill (15 Hz, 30+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.36 g	8.0 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		36%	22.5 ml/g		voluminous
MRa340A	A	1.28 g	MB2996	1.92 g	KOH	(90%)				ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.49 g	4.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		38%	8.2 ml/g		barely shiny
MRa340B	B	1.04 g	MB2996	1.56 g	KOH	(90%)				ball mill (15 Hz, 120 min)	yes	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.44 g	3.5 ml	1	black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		42%	7.9 ml/g		barely shiny
TB1155A	A	1.28 g	MB2986 (2% water)	1.54 g	KOH	(90%)	0.38 g	NaOH	(98%)	ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.69 g	4.0 ml		black/grey
		NaOH not fully ground!, ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		54%	5.8 ml/g		somewhat shiny
TB1155B	B	1.04 g	MB2986 (2% water)	1.25 g	KOH	(90%)	0.31 g	NaOH	(98%)	ball mill (15 Hz, 10 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.45 g	4.0 ml		black/grey
		NaOH not fully ground!, ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		43%	8.9 ml/g		somewhat shiny
MB3017A	A	1.28 g	MB3002 (2% water)	1.54 g	KOH	(90%)	0.38 g	NaOH	(98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.53 g	19.1 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		41%	36.3 ml/g		very voluminous
MB3017B	B	1.04 g	MB3002 (2% water)	1.25 g	KOH	(90%)	0.31 g	NaOH	(98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.33 g	12.2 ml	2	shiny black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		32%	36.7 ml/g		voluminous
MB3019A	A	1.28 g	MB3002 (2% water)	1.92 g	KOH	(90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.47 g	15.9 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		37%	33.9 ml/g		very voluminous
MB3019B	B	1.04 g	MB3002 (2% water)	1.56 g	KOH	(90%)				ball mill (15 Hz, 10+1 min)	no	N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.39 g	9.3 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		38%	23.7 ml/g		voluminous
MB3021A	A	1.28 g	MB3002 (2% water)	1.92 g	KOH	(90%)				mill (2 min)		N2	3 °C/min	800 °C	1 h	48 g	10% HCl	100 °C	N2 stream	0.52 g	7.3 ml	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size														(+ water)		(atm. Pressure)		41%	14.0 ml/g		
MB3021B	B	1.04 g	MB3002 (2% water)	1.56 g	KOH	(90%)				mill (2 min)		N2	3 °C/min	800 °C	1 h	39 g	10% HCl	100 °C	N2 stream	0.43 g	5.3 ml	Mostly 1	shiny black/grey
		ground biochar (brown powder) NOT compacted														(+ water)		(atm. Pressure)		42%	12.2 ml/g		

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3023A	A	1.08 g	MB2989	1.30 g KOH (90%) 0.32 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.28 g	16.5 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	26%	59.1 ml/g		very voluminous
MB3023B	B	0.80 g	MB2989	0.96 g KOH (90%) 0.24 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.25 g	13.5 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	31%	54.7 ml/g		very voluminous
MRa352A	A	1.28 g	MB3002 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.62 g	2.9 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	48%	4.7 ml/g		barely shiny
MRa352B	B	1.04 g	MB3002 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.53 g	2.8 ml	1	black/grey
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	51%	5.3 ml/g		barely shiny
MB353A	A	1.28 g	MB2991	1.92 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.53 g	2.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	41%	3.8 ml/g		barely shiny
MRa353B	B	1.04 g	MB2991	1.39 g KOH (90%) 0.17 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.44 g	3.5 ml	Mostly 1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	7.9 ml/g		barely shiny
TB1158A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	50 g 10% HCl	100 °C N2 stream	0.40 g	17.0 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	39%	42.3 ml/g		very voluminous
TB1158B	B	0.96 g	MB2991	1.37 g KOH (90%) 0.08 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.39 g	3.0 ml	Mostly 1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	40%	7.8 ml/g		barely shiny
TB1160A	A	1.04 g	MB2991	1.25 g KOH (90%) 0.31 g NaOH (98%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.46 g	17.0 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%	37.0 ml/g		very voluminous
TB1160B	B	1.04 g	MB2991	1.56 g KOH (90%)	ball mill (35 Hz, 10 min)	yes	N2	3 °C/min	800 °C	1 h	41 g 10% HCl	100 °C N2 stream	0.43 g	2.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	41%	4.6 ml/g		barely shiny
TB1162A	A	1.04 g	MB2989	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.44 g	4.0 ml	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	9.0 ml/g		
TB1162B	B	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.32 g	8.0 ml	Mostly 2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	31%	25.2 ml/g		
MB3026	B	1.04 g	MB3005 (2% water)	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.40 g	8.3 ml	1/2 mix	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	39%	20.6 ml/g		voluminous
MB3027	B	1.04 g	MB3005 (2% water)	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (3 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.28 g	18.0 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	27%	64.7 ml/g		very voluminous
MB3028	B	1.04 g	MB2989	1.56 g KOH (90%)	mill (10 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.42 g	5.1 ml	Mostly 1	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	40%	12.1 ml/g		

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MB3030	A	1.04 g	MB2989	1.25 g KOH (90%) 0.31 g NaOH (98%)	mill (10 min)	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.29 g	16.8 ml	2	shiny black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	28%	57.7 ml/g		very voluminous
TB1165	C	12.4 g	MB2992	18.5 g KOH (90%)	5 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	193 g 10% HCl	100 °C N2 stream	4.55 g	35.0 ml	1	dull black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	37%	7.7 ml/g		
TB1166	C	7.8 g	MB3006 (2% water)	9.41 g KOH (90%)	4 millings (3 min each)	yes	N2	3 °C/min	800 °C	1 h	210 g 10% HCl	100 °C N2 stream	3.25 g	65.0 ml	1/2 mix	shiny black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	41%	20.0 ml/g		
MB3036	B	1.04 g	MB2992 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.34 g	3.0 ml	1	black/grey
		water added to mill => 7-8% (calc), ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	32%	8.9 ml/g		barely shiny
TB1167	C	14.8 g	MB2994	22.3 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	200 g 10% HCl	100 °C N2 stream	6.40 g	50.0 ml	Mostly 1	shiny black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	43%	7.8 ml/g		
MB3038	B	1.04 g	MB2994	1.56 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.53 g	2.4 ml	Mostly 1	black/grey
		ground biochar (brown powder from TB1167) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	51%	4.5 ml/g		barely shiny
MRa373	A	1.04 g	MB2989 + H2O	1.56 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.43 g	3.5 ml	Mostly 1	shiny black/grey
		water added to biochar => 8% (KFT), ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	41%	8.2 ml/g		
MRa375	A	1.04 g	MB2996 + H2O	1.56 g KOH (90%)	mill (1+3 min)	yes	N2	3 °C/min	800 °C	1 h	40 g 10% HCl	100 °C N2 stream	0.51 g	3.2 ml	Mostly 1	black/grey
		water added to mill => 9% (calc), ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	49%	6.3 ml/g		barely shiny
MRa376	C	16.7 g	MB2998 + H2O	25.0 g KOH (90%)	blender, 10+80 sec	?	N2	3 °C/min	800 °C	1 h	189 g 10% HCl	100 °C N2 stream	7.6 g	60.0 ml	Mostly 1	shiny black/grey
		water added to blender => 7-8% (calc), rotating oven, powder not compacted									(+ water)	(atm. Pressure)	46%	7.9 ml/g		
MB3041	B	1.3 g	MB2996	1.3 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g	3.0 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	35%	6.5 ml/g		barely shiny
MB3043	C	16.5 g	MB3000	24.72 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	285 g 10% HCl	100 °C N2 stream	7.11 g	70.0 ml	Mostly 1	black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	43%	9.8 ml/g		barely shiny
MRa380	A	1.0 g	MB2996 + H2O	1.0 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.48 g	5.0 ml	1/2 mix	shiny black/grey
		water added to biochar => 11.5% (KFT), ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	46%	10.4 ml/g		
MB3046	C	15.6 g	MB3032	24.20 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	311 g 10% HCl	100 °C N2 stream	7.05 g	95.0 ml	*	shiny black/grey
		KOH not milled, rotating oven, powder not compacted									(+ water)	(atm. Pressure)	45%	13.5 ml/g		
MRa385	C	17.8 g	MRa367	26.6 g KOH (90%)	blender, 6x15 sec	no	N2	3 °C/min	800 °C	1 h	183 g 10% HCl	100 °C N2 stream	8.60 g	125.0 ml	Mostly	shiny black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	48%	14.5 ml/g	1/2 mix	
MRa386	A	2.1 g	MRa368	2.1 g KOH (90%)	mill (3 min)	yes	N2	3 °C/min	800 °C	1 h	60 g 10% HCl	100 °C N2 stream	0.65 g	3.7 ml	1	black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	32%	5.7 ml/g		barely shiny

Experiment	Oven	Qty	Lot	Base	grinding	homogeneous?	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Vol/Dens	Species	Appearance
MRa389A	C	17.2 g	MB3005/3008/3010 + H2O	25.8 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	178 g 10% HCl	100 °C N2 stream	8.2 g	70.0 ml	1	black/grey
		water added to blender => 5% (calc), rotating oven, powder not compacted									(+ water)	(atm. Pressure)	48%	8.5 ml/g		barely shiny
MRa389B	C	16.5 g	MB3005/3008/3010 + H2O	24.8 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	114 g 10% HCl	100 °C N2 stream	5.2 g	50.0 ml	1	black/grey
		water added to blender => 5% (calc), rotating oven, powder not compacted									(+ water)	(atm. Pressure)	31%	9.6 ml/g		somewhat shiny
MRa389C	C	15.2 g	MB3005/3008/3010 + H2O	22.9 g KOH (90%)	blender, 2x45 sec	?	N2	3 °C/min	800 °C	1 h	154 g 10% HCl	100 °C N2 stream	7.3 g	60.0 ml	1	black/grey
		water added to blender => 5% (calc), rotating oven, powder not compacted									(+ water)	(atm. Pressure)	48%	8.2 ml/g		barely shiny
MB3050	C	17.8 g	MB3044	17.8 g KOH (90%)	blender, 90 sec	yes	N2	3 °C/min	800 °C	1 h	125 g 10% HCl	100 °C N2 stream	8.1 g	40.0 ml	Mostly 1*	dull black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	46%	4.9 ml/g		
MB3051	C	16.3 g	MB3040	24.5 g KOH (90%)	blender, 4x45 sec	yes	N2	3 °C/min	800 °C	1 h	269 g 10% HCl	100 °C N2 stream	7.20 g	90.0 ml	Mostly 1	black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	44%	12.5 ml/g		somewhat shiny
MRa394	C	21.4 g	MB3045	21.4 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	148 g 10% HCl	100 °C N2 stream	7.2 g	40.0 ml	Mostly 1*	dull black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	34%	5.5 ml/g		
MB3054	C	16.0 g	MB3034	16.0 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	124 g 10% HCl	100 °C N2 stream	7.51 g	70.0 ml	*	dull black/grey
		Rotating oven, powder not compacted									(+ water)	(atm. Pressure)	47%	9.3 ml/g		
MRa395	A	1.3 g	MB3034	1.3 g KOH (90%)	blender, 90 sec	no	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.6 g	3.0 ml	Mostly 1*	dull black/grey
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	46%	5.0 ml/g		
MRa398A	C	25.6 g	MB3016/3018/3020	38.5 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	256 g 10% HCl	100 °C N2 stream	12.6 g	100.0 ml	1	dull black/grey
		Rotating oven, powder not compacted, powder turned dark during milling									(+ water)	(atm. Pressure)	49%	7.9 ml/g		
MRa398B	C	30.4 g	MB3016/3018/3020	45.6 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	303 g 10% HCl	100 °C N2 stream	14.2 g	90.0 ml	1	dull black/grey
		Rotating oven, powder not compacted, powder turned dark during milling									(+ water)	(atm. Pressure)	47%	6.3 ml/g		
MB3055	C	40.6 g	(various combined lots)	61.0 g KOH (90%)	blender, 2x45 sec	no	N2	3 °C/min	800 °C	1 h	573 g 10% HCl	100 °C N2 stream	19.4 g	105.0 ml	1	dull black/grey
		Rotating oven, powder not compacted, powder turned dark during milling									(+ water)	(atm. Pressure)	48%	5.4 ml/g		

Comments

- Yellow/red text for the lot of biochar show cases where the quality of the milled material was somewhat/significantly impaired by contact with air and/or moisture (leading to heating or colour change of the material, stickiness, clumping or similar changed in appearance)
- The information on homogeneity of the milled material is based on optical and acoustic evaluation